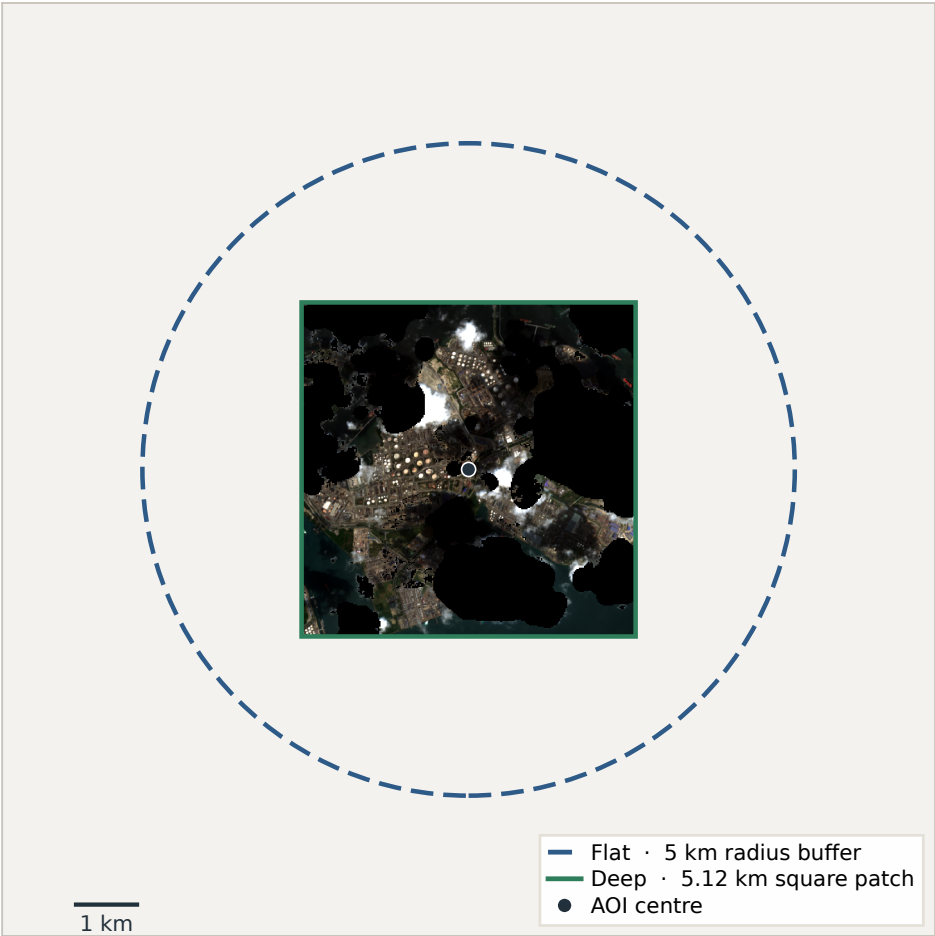


Figure 3.4 Worked example · Jurong Island (P004), forecast origin 14 March 2025

Latest eligible observations under the source-specific availability lags.

(a) Spatial supports · January 2025 Sentinel-2 composite

Under the locked-lag rule, the February 2025 composite was not yet eligible at this forecast origin. January inputs were eligible from 15 February and carried across the four-week input window.



(b) Remote-sensing representations

Flat	Deep
Sentinel-2 + VIIRS scalar anomalies	Sentinel-2 representation
5 km buffer · five site-level scalars	Sentinel-2 patch → resize to 224 × 224
	→ frozen Prithvi → mean pooling
	→ 1024-d embedding
NDVI anomaly	+0.56
NDWI anomaly	-0.40
NDBI anomaly	-0.93
BSI anomaly	-0.76
NTL anomaly	+0.36

e₁

e₂

e₃

e₄

e₅

...

e₁₀₂₄

(c) Shipping graph inputs around Jurong

GFW voyage edges: week ending 28 February
PortWatch tanker calls: week ending 7 March.

Dynamic GFW voyage edges	Fixed corridor edge
16 directed lanes · 257 voyages busiest four shown	Jurong Malacca
Origin → destination	Voyages
Ningbo → Jurong	58
Jurong → Ningbo	45
Fujairah → Jurong	26
Jurong → Fujairah	23

Selected node attribute

Jurong

PortWatch tanker calls · 520